MEMO Final - FSK116 Normal Exam 21 June 2023 ver4 GS

Physics for Engineers (University of Pretoria)

Department of Physics, FSK 116, June Examination

NEATLY FILL-IN YOUR DETAILS IN THE BOXES BELOW!
Do not write anything else inside the dashed box – not even your signature!

Surname and Names	Student Number

For Official Use ONLY. Apply ✓ if cheat-sheet is compliant (Leave empty if not. **Students who apply ticks here, will lose whatever points they would otherwise earn for their cheat sheet!**

Examiners: Mr MJ Legodi
Moderator: Prof WE Meyer

Date: 21 June 2023
Time: 15h00 to 18h00

Total 71

The following specific rules apply to this Test/Examination

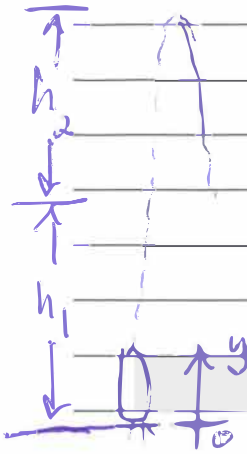
1. Fill in your details at the bottom of every page.
2. Answer all questions **ONLY** in the spaces provided. If you need more space, use the "overflow pages" at the back of the question paper and clearly indicate where each question continues. Ask for more overflow pages from invigilators if you require more. No work outside these guidelines will be marked!
3. Pocket calculators may be used, but programmable calculators, cellphones, earphones, electronic watches and other such devices are prohibited.
4. This paper may not be taken out of the test/examination venue. It must be handed in to the chief invigilator, before you leave the venue.
5. Work written in pencil will not be marked. **Write in black permanent pen**.
6. Provide labelled sketches, including co-ordinate systems, free-body diagrams, final answers with correct significant figures and units, etc. Marks will be lost, otherwise.
7. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant Physics. Only substitute numerical values in the final step of calculations! A correct answer does not mean that you will receive full marks.
8. Units must be used when values/numbers are substituted, else marks will be deducted.
9. Extra points will be awarded for high quality or elegant answers.

Constants, conversion factors and some formulae that may be useful

$g = 9.80\ \text{m/s}^2$ $L_F = 3.33\times10^5$ J/kg $L_V = 2.26\times10^6$ J/kg $c_{water} = 4186$ J/(kg °C) $c_{steam} = 2021$ J/(kg °C)
$m_{Earth} = 5.9722\times10^{24}$ kg

QUESTION 1 [9]

1.1 A rocket, initially at rest, is fired vertically with an upward acceleration of 10 m/s^2. At an altitude of 0.50 km, the engine of the rocket cuts off. What is the maximum altitude it achieves? (4)



$a_{y1} = 10\,m/s^2$ and $a_{y2} = -g = -9.80\,m/s^2$

$v_{y1f}^2 = v_{y1i}^2 + 2a_{y1}(y_{1f} - y_{1i})$

$v_{yf} = \sqrt{2(10\,m/s^2)(500\,m)} = 100\,m/s$

4

h_2: $v_{y2f} = v_{y2i} + a_{y2}t$

$0 = 100\,m/s - (9.80\,m/s^2)t$

$\therefore t = \dfrac{100\,m/s}{9.80\,m/s^2} = 10.20408163\,s$

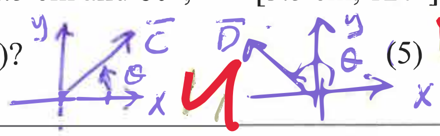
h_2: $y_{2f} - y_{2i} = v_{y2i}t + \frac{1}{2}a_{y2}t$

$= (100\,m/s)(10.2041\,s) - \left(\dfrac{9.80\,m/s^2}{2}\right)(10.2041\,s)^2$

$= 510.2041\,m$

Max height $= h_1 + h_2 = 1.010204\,m = 1.0\,km$

1.2 If $\vec{C}$ = [2.5 cm, 80°], i.e., the magnitude and direction of $\vec{C}$ are 2.5 cm and 80°, $\vec{D}$ = [3.5 cm, 120°], and $\vec{E} = \vec{D} - 2\vec{C}$, what is the direction of $\vec{E}$ (to the nearest degree)? (5)



$C_x = C\cos\theta = 0.43412\,cm$; $D_x = D\cos\theta = -7/4\,cm$

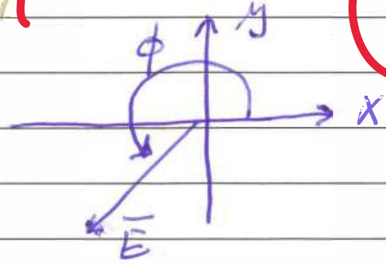
$C_y = C\sin\theta = 2.46202\,cm$; and $D_y = D\sin\theta = \left(\frac{7}{4}\right)\sqrt{3}\,cm$

From $\vec{E} = \vec{D} - 2\vec{C} = [-1.75 - 2(0.43412)]\,cm\,\hat{i} + [3.0310889 - 2(2.462019)]\,cm\,\hat{j}$

$= (-2.6182\hat{i} - 1.8929498\hat{j})\,cm$

5

$\tan\phi = \dfrac{E_y}{E_x}$



$= 0.722985$

$\therefore \phi = 35.866° + 180°$

$= 216°$

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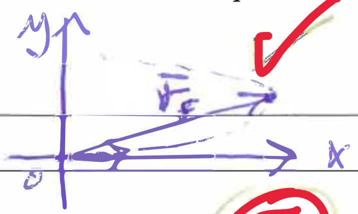
QUESTION 2 [25]

2.1 A particle starts from the origin at t = 0.0 s with a velocity of 6.0 $\hat{\mathbf{i}}$ m/s and moves in the xy plane with a constant acceleration of (–2.0 $\hat{\mathbf{i}}$ + 4.0 $\hat{\mathbf{j}}$) m/s^2. At the instant the particle achieves its maximum positive x coordinate, how far is it from the origin? (5)



$\bar{v}_i = \bar{v}_{xi} = 6.0\hat{i}$ m/s

$\bar{a} = -2.0\hat{i} + 4.0\hat{j}$ m/s^2

We need to determine x_f and y_f at the turning point in the x-axis.

5

Determine t from $v_{xf} = v_{xi} + a_x t \to t = \frac{v_{xf} - v_{xi}}{a_x} = \frac{0 - 6.0\,m/s}{-2.0\,m/s^2} = 3.0\,s$

$r_x = x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$

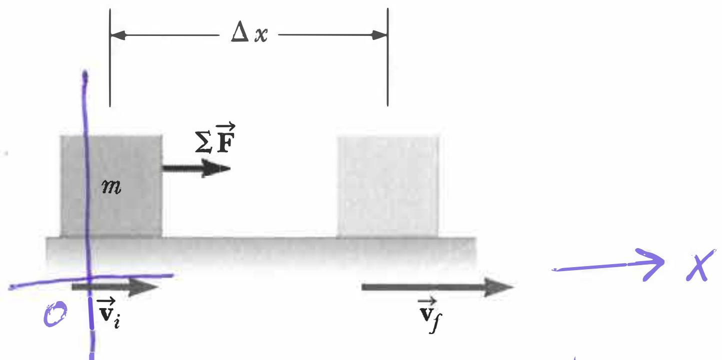
$= (6.0\,m/s)(3.0\,s) + \frac{1}{2}(-2.0\,m/s^2)(3.0\,m/s^2)^2 = 9.0\,m$

$r_y = y_f - y_i = v_{yi}t + \frac{1}{2}a_y t^2$

$= 0 + \frac{1}{2}(4.0\,m/s^2)(3.0\,s)^2 = 18\,m$

$|r_f| = \sqrt{r_x^2 + r_y^2} = \sqrt{(9.0\,m)^2 + (18\,m)^2} = 20\,m$ from (x_i, y_i) with $x_i = 0$ and $y_i = 0$. Alternative get t from $\frac{dx}{dt} = 0$

2.2 A net force, ΣF, is acting on a block of mass, m, and accelerates it from an initial velocity, $\boldsymbol{v}_i$, to a final velocity $\boldsymbol{v}_f$. The acceleration, $\boldsymbol{a}$, is uniform and the block undergoes a displacement Δx in the x axis over a flat and frictionless surface. Show that the work done by the net force is equal to the change in kinetic energy of the block. That is, derive the Work-Kinetic Energy Theorem. Show all your steps and reasoning. (5)



Consider the x-component motion only, to simplify.

Work done by an external force is defined as:

$W_{ext} = \int \bar{F}_{net} \cdot d\bar{r} \to \int F\hat{i} \cdot dx\hat{i}$ in the x-axis.

$= \int_{x_1}^{x_2} \Sigma F\,dx = \int_{x_i}^{x_f} ma\,dx$

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$$\therefore W_{ext} = m \int_{x_i}^{x_f} \left(\frac{dv}{dt}\right) dx$$

$$= m \int_{x_i}^{x_f} \left(\frac{dv}{dx}\right)\left(\frac{dx}{dt}\right) dx \quad \text{since } v(x(t)) \text{ by the chain rule}$$

Bonus

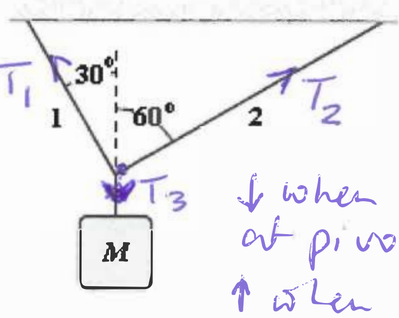
$$\Rightarrow W_{ext} = m \int_{v_i}^{v_f} v\,dv \quad \text{after simplification and change of variables.}$$

$$= m \left[\frac{v^2}{2}\right]_{v_i}^{v_f}$$

5

$$= \tfrac{1}{2} m v_f^2 - \tfrac{1}{2} m v_i^2 = \Delta K,$$ which establishes the Work-Kinetic energy Theorem required.

2.3 If $M = 6.0$ kg, what is the tension in string 1? Show all the details and support all your claims or assertions with physics arguments!



$\sum F_y = ma = 0$

T_3, M, mg

y, x, O

$-\frac{1}{2}$

T_1, T_2, T_3

↓ when looking at pivot point

↑ when analysing m

$T_1 \cos 30°$ $\sum F_y = 0$

$T_2 \cos 60°$

pivot

T_3

(4)

4

$\sum F_x = ma = 0$

$T_1 \sin 30° \leftarrow$ pivot $\rightarrow T_2 \sin 60°$

$T_3 - Mg = 0$ from $\sum F_y = 0$

$\Rightarrow T_3 = (6.0\text{ kg})(9.80\text{ m/s}^2)$

$= 58.8\text{ N}$

At pivot point $\sum F_x = m a_x = 0$

$\Rightarrow T_1 \sin 30° = T_2 \sin 60°$

$\Rightarrow T_2 = T_1 \left(\frac{\sin 30°}{\sin 60°}\right)$

Also, $\sum F_y = 0$

$\Rightarrow T_1 \cos 30° + T_2 \cos 60° = T_3$

$\therefore T_1 \cos 30° + \left(T_1 \frac{\sin 30°}{\sin 60°}\right) \cos 60° = 58.8\text{N}$

$\therefore T_1 \left(\cos 30° + \frac{\sin 30°}{\cos 60°}\right) = 58.8\text{N}$

$\Rightarrow T_1 = (58.8\text{N}/1.154700)$

$T_1 = 51\text{ N}$

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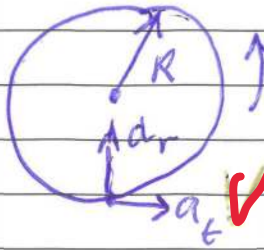
2.4 A carnival Ferris wheel has a 15.0 m radius and completes 5 turns about its horizontal axis every minute. What is the total acceleration of a passenger at his lowest point during the ride?

HINT: Each complete turn equals 2π radians. (5)



$R = 15.0\,m$

$\omega = \dfrac{5\ rev}{1\,min}$

$= \dfrac{5\ rev \times \left(\dfrac{2\pi\,rad}{1\,rev}\right)}{1\,min\left(\dfrac{60\,s}{1\,min}\right)}$

$= \dfrac{\pi}{6}\ rad/s$

But $a_{Total} = a_t + a_r$

$a_{Tot} = 0 + \dfrac{v^2}{R}$

$= \omega^2 R$

$= \left(\dfrac{\pi}{6}\right)^2 \cdot 15.0\,m/s^2$

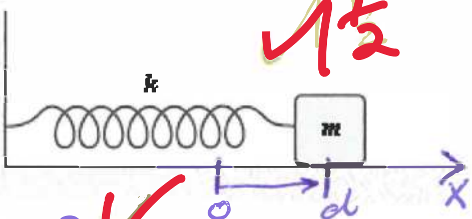
$= 4.11\ m/s^2$ or $4\ m/s^2$ if the 5 rev and 1 minute are treated as infinitely known.

The direction is to the origin.

5

2.5 The block shown is released from rest when the spring is stretched a distance d. If $k = 50$ N/m, $m = 0.50$ kg, $d = 10$ cm, and the coefficient of kinetic friction between the block and the horizontal surface is equal to 0.25, determine the speed of the block when it first passes through the position for which the spring is unstretched. (6)

System: Block, Spring, Surface.



$1\frac{1}{2}$

$d = 10\,cm = 0.10\,m$

$k = 50\,N/m$

$m = 0.5\,kg$

$\mu_k = 0.25$

6

$\Delta K + \Delta U_s + \Delta E_{int} = 0$

Consider instances (i) when spring is stretched to d, and (f) when the spring is at $x = 0$, its equilibrium point.

$\therefore (K_f - K_i) + \left(\frac{1}{2}kx_f^2 - \frac{1}{2}kx_i^2\right) + f_k d = 0$

$\left(\frac{1}{2}mv_f^2 - 0\right) + \left(0 - \frac{1}{2}kd^2\right) + f_k d = 0$ with $f_k = \mu_k F_N$ and $F_N = mg$

$\therefore \frac{1}{2}mv_f^2 = \frac{1}{2}kd^2 - \mu_k F_N d$

$\therefore v_f = \sqrt{\dfrac{k}{m}d^2 - 2\mu_k g d}$

$1\frac{1}{2}$

$= \sqrt{\left(\dfrac{50\,N/m}{0.5\,kg}\right)(0.10\,m)^2 - 2(0.25)(9.80\,m/s^2)(0.10\,m)}$

$= 0.714\ m/s = 71\ cm/s$

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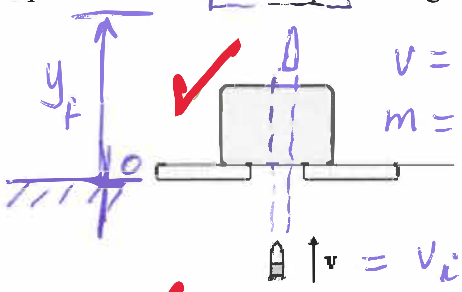
QUESTION 3 **(20)**

3.1 A 10 gram bullet moving 1 000 m/s strikes and passes through a 2.0 kg block initially at rest, as shown. The bullet emerges from the block with a speed of 400 m/s. To what maximum height will the block rise above its initial position? Show all steps and arguments/reasoning. (10)



$v = 400\,m/s$
$m = 2.0\,kg$

$= V_i = 1000\,m/s$
$m_b = 10\,g = 0.010\,kg$

Energy System: Block, Earth.

$\Delta K + \Delta U_g = 0$ OR

$(K_f - K_i) + (mgy_f - mgy_i) = 0$

$(0 - \frac{1}{2}mv^2) + (mgy_f - 0) = 0$

10

$mgy_f = \frac{1}{2}mv^2$

$y_f = \frac{v_B^2}{2g}$ with $v_{f,B}$ determined below

$= \frac{(3.0\,m/s)^2}{2(9.80\,m/s^2)} = 459\,mm = 46\,cm$

Bullet-Block momentum interaction

$m_b v_{b,i} + m_2(v_{B,i} = 0) = m_b v_{f,b} + m_B v_{f,B}$

$v_{f,B} = \frac{m_b v_{b,i} - m_b v_{f,b}}{m_B}$

$= \frac{m_b(v_{b,i} - v_{f,b})}{m_B}$

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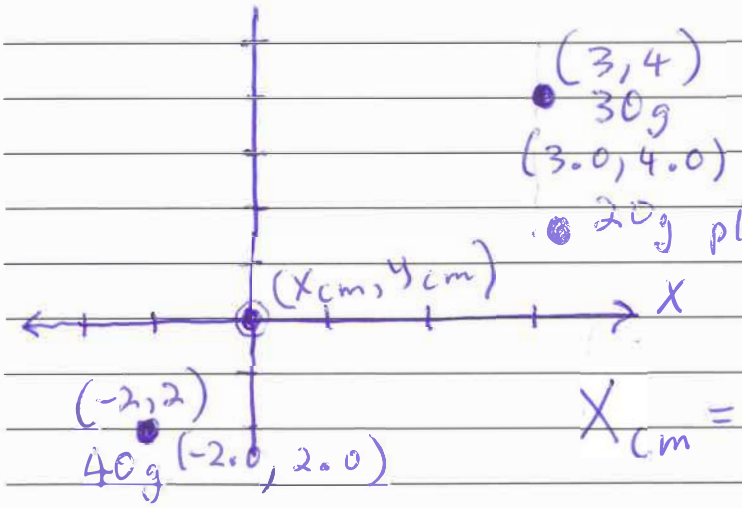
$$V_{f,B} = \frac{(0.010\,kg)(1000\,m/s - 400\,m/s)}{2.0\,kg}$$

$$= 3.0\,m/s$$

3.2 Three particles are placed in the *xy* plane. A 30 g particle is located at (3.0, 4.0) m, and a 40 g particle is located at (–2.0, –2.0) m. Where must a 20 g particle be placed so that the center of mass of the three-particle system is at the origin? (4)



4

$$X_{cm} = \frac{\sum m_i x_i}{\sum m_i} \quad \text{and} \quad y_{cm} = \frac{\sum m_i y_i}{\sum m_i}$$

$$\sum m_i = 30g + 40g + 20g = 90g$$

$$\therefore X_{cm} = 0 = \frac{\sum m_i x_i}{\sum m_i} = \frac{(30g)(3.0m) + (40g)(-2.0m) + (20g)(x)}{90g}$$

$$\therefore x = -0.5\,m$$

$$\text{And } y_{cm} = 0 = \frac{\sum m_i y_i}{\sum m_i} = \frac{(30g)(4.0m) + (40g)(-2.0m) + (20g)(y)}{90g}$$

$$\therefore y = -2.0\,m$$

$\therefore$ For 20g mass, the location should be $(-0.5m, -2.0m)$

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3.3 The linear density of a rod, in g/m, is given by $\lambda(x) = 40.0 + 30.0x$. The rod extends from the origin to $x = 0.400$ m. What is the location of the center of mass of the rod? Start off by calculating the total mass of the rod. (6)

We start off with the total mass calculation

$$M = \int_0^L \lambda\, dx \rightarrow M = \int_0^{0.400} (40.0 + 30.0x)\, dx \quad \left[\left(\frac{g}{m}\right) m\right] \text{ units}$$

$$= \left[40.0x + \frac{30.0}{2}x^2\right]_0^{0.400} \qquad g = 18.4\ g$$

6

$$x_{cm} = \int_0^{M} \left(\frac{1}{M}\right) x\, dm$$

$$= \int_0^{0.400} \frac{1}{M}\cdot \lambda \cdot x\, dx$$

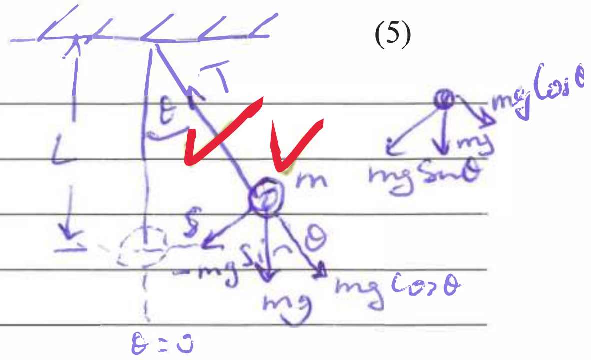
$$= \frac{1}{18.4\,g}\int_0^{0.400} (40.0x + 30.0x^2)\, dx \quad [g\,m]$$

$$= \frac{1}{18.4\,g}\left[\frac{40.0x^2}{2} + \frac{30.0}{3}x^3\right]_0^{0.400} \quad [g\,m]$$

$$= 0.209\ m$$

QUESTION 4 **[17]**

4.1 When θ is small, the motion of a simple pendulum can be modeled as simple harmonic motion (SHM) about the $\theta = 0°$ equilibrium position. Consider a mass m attached to the ceiling through a massless and non-stretchable cord. The force of gravity and the tension T are the only forces acting on the pendulum. If, at some instant in time, the pendulum is given a small displacement θ from the equilibrium position, derive an expression for the second order ordinary differential equation (for θ(t)) governing the motion of the pendulum. (5)



From $\Sigma F = ma$, the tangential component of $F = mg$ is the restoring force, forcing m to its $\theta = 0$ position.

$$-mg\sin\theta = ma = m\frac{d^2s}{dt^2}$$

But $s = L\theta$ and for small angles $\sin\theta \approx \theta \Rightarrow -mg\theta = m\frac{d^2(L\theta)}{dt^2} \rightarrow L\frac{d^2\theta}{dt^2} = -g\theta$

$$\therefore \frac{d^2\theta}{dt^2} = -\left(\frac{g}{L}\right)\theta$$

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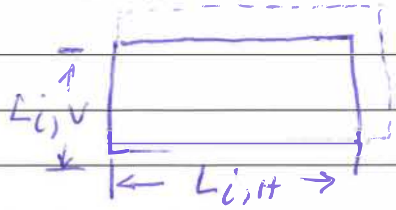
Upon re-defining $\frac{g}{L} = \omega^2$, we can write the equation of motion as $\frac{d^2\theta}{dt^2} = -\omega^2\theta$ ✓

4.2 What is the change in area (in cm^2) of a 60.0 cm by 150 cm automobile windshield when the temperature changes from 0.00 °C to 36.0 °C. The coefficient of linear expansion of this glass is 9.00×10^{-6} /°C, and you may assume the windscreen to be thin and rectangular in shape. (4)



$A_0 = L_{i,H} \times L_{i,V}$
$= 150\text{ cm} \times 60\text{ cm}$
$= 9.0 \times 10^3\text{ cm}^2$ ½

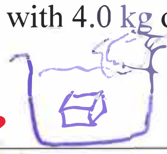
$\Delta L_H = L_{i,H}\,\alpha\,\Delta T$ and $\Delta L_V = L_{i,V}\,\alpha\,\Delta T$
$= (150\text{ cm})(9.0\times10^{-6}/°C)(36.0°C - 0.0°C)$ and $\Delta L_V = (60\text{ cm})(9.0\times10^{-6}/°C)(36.0°C)$
$= 48.6\times10^{-3}\text{ cm}$ $= 19.4\times10^{-3}\text{ cm}$

$\therefore A_{new} = (L_{i,H} + \Delta L_H)(L_{i,V} + \Delta L_V)$ ✓ OR ✓
$= (150.0486\text{ cm})(60.01944\text{ cm})$
$= 9.00583294\times10^3\text{ cm}^2$ ½

4

$\Delta A_{rea} = A_{new} - A_0$ ✓
$= 5.83294\text{ cm}^2$
$= 5.8\text{ cm}^2$ ✓

4.3 If 25 kg of ice at 0.0 °C is combined with 4.0 kg of steam at 100 °C, what will be the final equilibrium temperature (in °C) of the system? (8)



✓ Bonus

Suppose the ice-steam mixture is isolated
✓ $\Delta Q = 0$ for isolated sys.
OR ✓ $Q_{hot} = Q_{cold}$

$m_s = 4.0\text{ kg}$
$T_{i,ice} = 0.0°C$
$T_{i,steam} = 100°C$
$C_w = 4186\text{ J/(kg °C)}$
$L_F = 3.33\times10^5\text{ J/kg}$
$L_V = 2.26\times10^6\text{ J/kg}$

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$-(Q_{s\to w} + Q_{Hot\ water}) = Q_{ice\to water} + Q_{cold\ water}$

$-[(-L_v m_s) + m_{HW} C_w(T_f - T_{i,HW})] =$ 3pts.

$L_F m_{ice} + m_{ice} C_w (T_f - T_{i,cold\ water})$

neg. pos

$L_v m_s - m_{HW} C_w T_f + m_{HW} C_w T_{i,HW} =$

$L_F m_{ice} + m_{ice} C_w T_f - m_{ice} C_w T_{i,cold\ water}$

$L_v m_s + m_{HW} C_w T_{i,HW} - L_F m_{ice} + m_{ice} C_w T_{i,cold\ water}$

$= (m_{ice} C_w + m_{HW} C_w) T_f$

$$\therefore T_f = \frac{L_v m_s - L_F m_{ice} + m_{HW} C_w T_{i,HW} + m_{ice} C_w T_{i,cw}}{(m_{ice} C_w + m_{HW} C_w)}$$

$$= \frac{(2.26\times10^6)(4.0) - (3.33\times10^5)(25) + (4.0)(4186)(100) + (0^\circ C)}{(25 + 4.0)(4186)}$$

after dropping units

OR

$$= \frac{9.04\times10^6 - 8.325\times10^6 + 1.6744\times10^6 + 0}{121.894\times10^3}\ ^\circ C$$

$= 19.6830\ ^\circ C$

$= 20\ ^\circ C$

value, units, sig Fig

GOOG LUCK

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